A COLLECTION OF OPEN SETS CLOSED UNDER FINITE INTERSECTIONS AND CONTAINING THE MEMBERS OF AN OPEN COVER IS A BASIS

THE STACKS PROJECT AUTHORS

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Source reference. Original definition definition-base: sources/stacks-topology/pinned-original-source.tex, source lines 499–499. Navigation pointer only: the counted text refers to the Stacks item definition-base, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a set and let $\mathcal{B}$ be a collection of subsets. Assume that $X = \bigcup_{B \in \mathcal{B}} B$ and that given $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$ there is a $B_3 \in \mathcal{B}$ with $x \in B_3 \subset B_1 \cap B_2$. Then there is a unique topology on X such that $\mathcal{B}$ is a basis for this topology.*

Proof. Let $\sigma(\mathcal{B})$ be the set of subsets of X which can be written as unions of elements of $\mathcal{B}$. We claim $\sigma(\mathcal{B})$ is a topology. Namely, the empty set is an element of $\sigma(\mathcal{B})$ (as an empty union) and X is an element of $\sigma(\mathcal{B})$ (as the union of all elements of $\mathcal{B}$). It is clear that $\sigma(\mathcal{B})$ is preserved under unions. Finally, if $U, V \in \sigma(\mathcal{B})$ then write $U = \bigcup_{i \in I} U_i$ and $V = \bigcup_{j \in J} V_j$ with $U_i, V_j \in \mathcal{B}$ for all $i \in I$ and $j \in J$. Then

$$U \cap V = \bigcup\nolimits_{i \in I, j \in J} U_i \cap V_j$$

The assumption in the lemma tells us that $U_i \cap V_j \in \sigma(\mathcal{B})$ hence we see that $U \cap V$ is too. Thus $\sigma(\mathcal{B})$ is a topology. Properties (1) and (2) of Definition definition-base are immediate for this topology. To prove the uniqueness of this topology let $\mathcal{T}$ be a topology on X such that $\mathcal{B}$ is a base for it. Then of course every element of $\mathcal{B}$ is in $\mathcal{T}$ by (1) of Definition definition-base and hence $\sigma(\mathcal{B}) \subset \mathcal{T}$. Conversely, part (2) of Definition definition-base tells us that every element of $\mathcal{T}$ is a union of elements of $\mathcal{B}$, i.e., $\mathcal{T} \subset \sigma(\mathcal{B})$. This finishes the proof. □

Date: stacks-project:topology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.